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Local structural aspects of the orthorhombic to pseudo-cubic phase transformation in $La_{1-x}Ca_xMnO_3$

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Abstract

The local and intermediate structure of $La_{1-x}Ca_xMnO_3$ has been studied in the orthorhombic (O) and pseudo-cubic (O′) phases, both as a function of temperature ($x = 0.00$), and as a function of Ca content (at 310 K). Neutron powder diffraction-based high real space resolution atomic pair distribution function (PDF) analysis shows that the Jahn Teller distortion of the MnO_6 octahedra *persists locally* deep in the O′ phase in both cases studied, contrary to the average crystallographic view. The O to O′ structural phase transformation does have a local structural signature evidenced as a dramatic change in a PDF peak at 10.3 Å sensitive to the rotations of the MnO_6 octahedra, providing further evidence that the nature of the transformation is orbital order to disorder regardless of whether the phase boundary is crossed as a function of T or x.
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The manganite-based perovskite $La_{1-x}Ca_xMnO_3$ family has been the subject of significant interest due to the complex interplay of orbital, electronic, and magnetic degrees of freedom [1]. The undoped endmember $LaMnO_3$ at low temperature is an A-type antiferromagnet with long-range ordered, Jahn–Teller (JT) distorted, MnO_6 octahedral structural units, comprising four shorter and two longer Mn–O bonds, and has orthorhombic (O) symmetry (s.g. Pbnm) [2]. When the temperature is increased above $T_{JT}\sim750$ K [3], structural transformation occurs to a pseudo-cubic phase (O′), without space group change. This JT-transformation is accompanied by a loss of long-range orbital order, and JT-distortions are virtually eliminated crystallographically. Ca-doping abruptly lowers the O–O′ phase transition temperature, which drops to about 310 K at $x = 0.12$, and to 35 K at $x = 0.2$ [3].

While the O′-phase is extremely important due to the development of ferromagnetism and colossal magnetoresistance, there is disagreement about its exact nature [4–6]. The JT-transition in the undoped endmember was found to be of orbital order to disorder character [5,7]. In addition, recent local structural study showed that JT-distortion survives locally in full magnitude, and that the ground-state-like orbital order remains within clusters ~16 Å in diameter [8].

Here we extend these results by comparing the local structural behavior as the O–O′ transition is crossed both vs temperature at fixed doping, and vs doping at fixed temperature. The local structural information is accessed by means of the real-space atomic pair distribution function (PDF) analysis of the neutron diffraction data collected at NPDF diffractometer at LANSCE (fixed composition), and at GEM diffractometer at ISIS (fixed temperature). The temperature dependence of the $x = 0.00$ sample was carried out at 14 temperatures from 300 up to 1150 K, while the doping dependence used samples with

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various Ca content from $x = 0.0$ up to $x = 0.5$. The experimental PDF, $G(r)$, is obtained through a sine Fourier transform of the total scattering structure function, $S(Q)$. The radial distribution function (RDF), $R(r)$, relates to $G(r)$ through a series of simple arithmetic operations. Both functions are peaked at distances separating pairs of atoms. The PDFs presented here used high resolution total scattering data up to a very high value of momentum transfer ($Q_{MAX} = 35\,\text{Å}^{-1}$). Sample preparation, experimental procedure, data processing methodology, and the definitions of various PDF variants used in this work are described in detail elsewhere [9,10]. Structural modeling was carried out using the refinement program PDFFIT [11].

Fig. 1 shows a series of experimental RDFs, focusing on the low-r region to illustrate the evolution of the JT-distortion in the two cases. Temperature in Fig. 1(a) and x in 1(b) both increase from top to bottom. The vertical dashed lines indicate positions of the short and long Mn–O bonds in the ground state of undoped $LaMnO_3$. The two RDF peaks are negative due to the negative neutron scattering length of Mn. The local JT-distortion clearly survives in the pseudo-cubic O′ phase as the phase boundary (indicated by dashed RDF profiles) is crossed from the O to O′ phase both at fixed x and fixed T, suggesting that the transformation has the same orbital order to disorder origin in both cases.

Further experimental evidence pointing towards the same origin of the transformation for the two cases is presented in Fig. 2. This shows an abrupt change observed in the 10.3 Å RDF peak height upon phase transformation, a feature that remained unnoticed in the earlier PDF studies of this system. The qualitative similarity of the RDF features in the two cases is striking.

In order to investigate the origin of this abrupt change in the 10.3 Å peak, a qualitative analysis was carried out. This can be done by systematically changing different sublattices in the models and seeing which of these changes affect the 10.3 Å peak in the manner observed in the data. Selected results are shown in Fig. 3. The solid lines show the calculated PDFs from fully converged models fitting

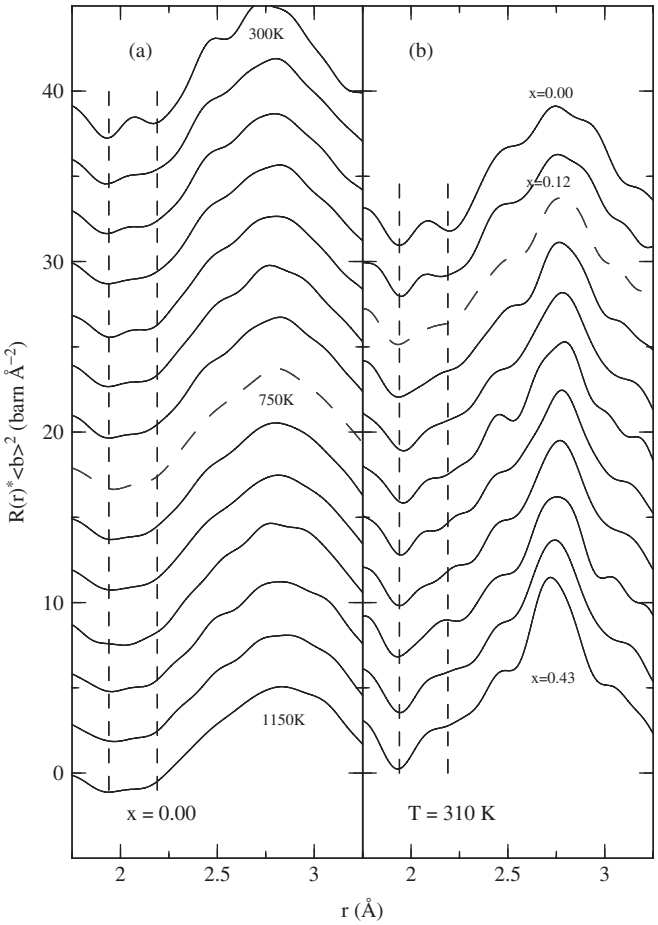


Fig. 1. Evolution of the JT-distortion in $La_{1-x}Ca_xMnO_3$ with temperature at fixed composition (a) and with Ca content at fixed temperature (b). Both temperature and doping concentration increase from top to bottom. The experimental $R(r)$ profiles are offset for clarity. Thick dashed profiles in both panels correspond to the O to O′ transformation temperature and Ca concentration [3]. Vertical dashed lines indicate the position of long and shorter Mn-O bondlengths in the MnO_6 octahedron. The fixed composition and temperature are as indicated in the panels, as well as selected variable parameters.

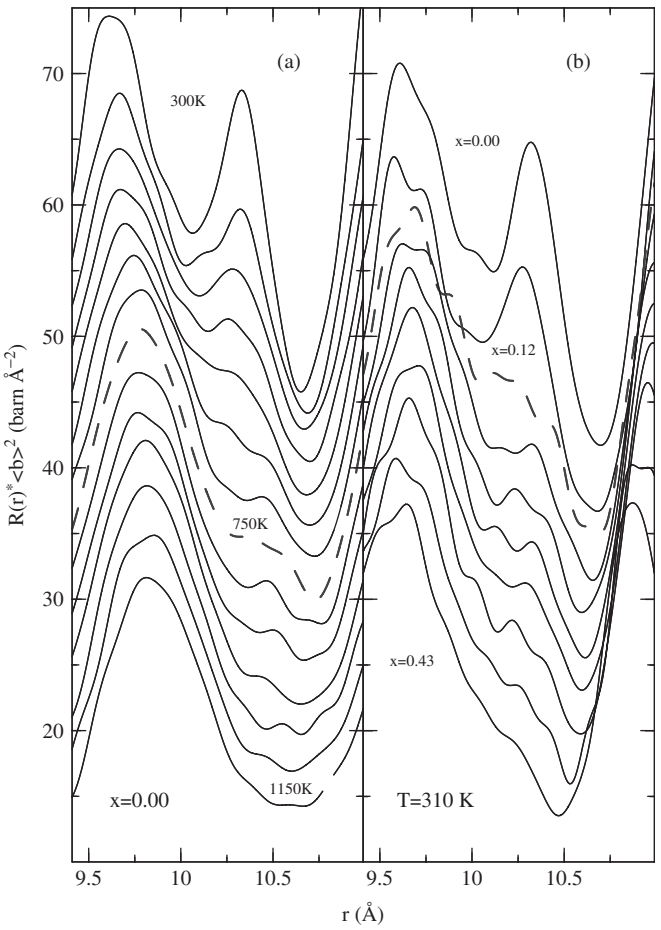


Fig. 2. Evolution of the 10.3 Å peak in $R(r)$ with temperature at fixed composition (a) and with Ca content at fixed temperature (b). The qualitative behavior of the peak across the O to O′ transformation boundary is the same in the two cases.

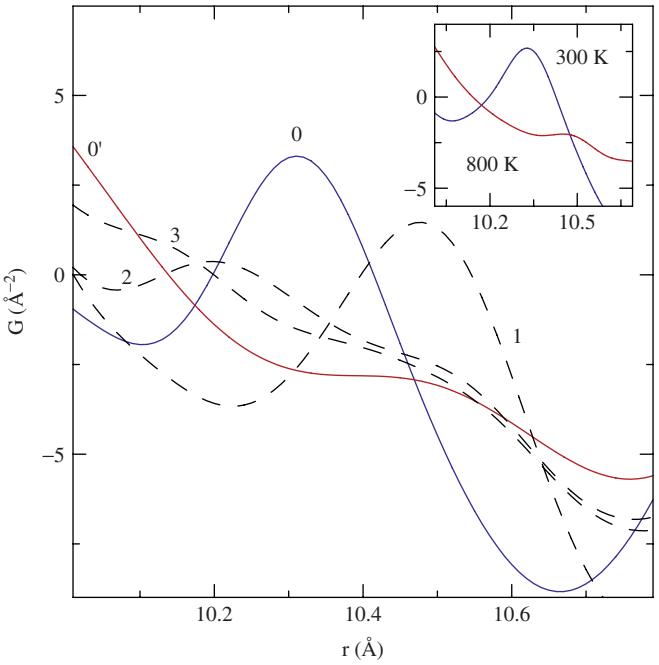


Fig. 3. Qualitative analysis of the nature of the abrupt 10.3 Å peak change upon O to O′ transformation based on refined structures. The inset shows the data at 300 K (O-phase) and 800 K (O′-phase) for $x = 0.00$. See text for details.

the orthorhombic (O) and pseudo-cubic (O′) datasets. These reproduce the data (inset) very well. Changes in the PDF due to rearrangement in La, O1, and O2 sublattices alone are shown as dashed lines labeled 1, 2, and 3 respectively. Curves 2 and 3 capture the peak disappearance indicating that it originates from the oxygen sublattice, presumably related to the rotational degrees of freedom. An orbital order-to-disorder transition will affect strongly the oxygen sublattice. To quantify the result of this study, Fig. 4 shows the JT-parameter [12] (solid circles), extracted from the PDF refinements, representing an RMS measure of the JT-distortion. This changes monotonically through the transition for both modes of crossing the phase boundary. On the other hand, the 10.3 Å RDF peak height extracted from the data (open squares), which is sensitive to the MnO_6 octahedral rotations, changes abruptly at the transition. The above two quantities were normalized to their values at 300 K for $x = 0.00$, and, in addition, a monotonic doping dependence of the RDF peak height has been extracted out [10].

In summary, we have demonstrated that the local JT-distortion is not sensitive to the O–O′ transformation, neither when crossed as a function of temperature at fixed composition, nor as a function of Ca content at fixed

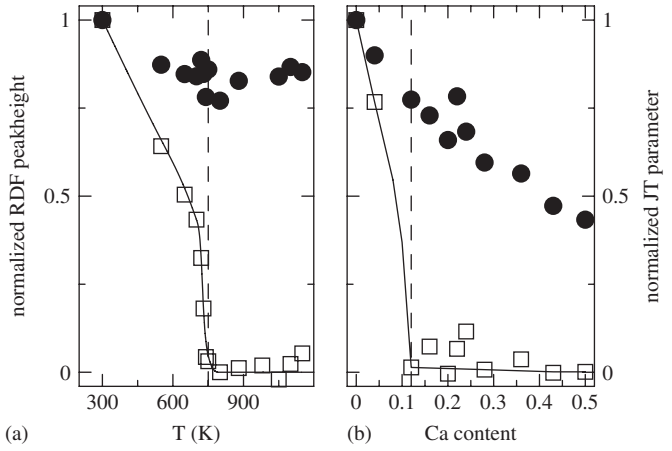


Fig. 4. Normalized 10.3 Å RDF peak height (open squares) and normalized JT-parameter (solid circles) as a function of temperature at fixed composition (a) and as a function of Ca content at fixed temperature (b). Solid lines are guides to the eye. Vertical dashed lines indicate O to O′ transformation.

temperature. Features related to the octahedral rotations, however, change abruptly at the transition, emphasizing the orbital order to disorder nature of the transition irrespective of the way the phase boundary is crossed.

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